

NexRay AI

Medical Assistant Platform • Clinical Decision Support Report

10 August 2026 | 08:44 PM

Session ID: 57

Patient: Paul

Modules: X-Ray Analysis

X-Ray Analysis

Detected Region: Chest (Thorax)

No significant findings detected.

■ **Urgency Level: Urgent — Same day attention required**

Recommended Tests

- Sputum smear microscopy (ZN stain) and GeneXpert MTB/RIF for TB exclusion
- Blood culture and complete blood count with differential
- Chest X-ray with lateral view for better localization
- Echocardiography if heart failure suspected
- Procalcitonin or CRP to differentiate bacterial from viral etiology

Suggested Treatment

- Initiate broad-spectrum antibiotics (Amoxicillin-clavulanate or Ceftriaxone) pending culture results if pneumonia suspected
- Oxygen therapy to maintain SpO2 >94%
- Supportive care including hydration and fever management
- Do NOT start anti-TB therapy until TB is confirmed - await GeneXpert results

Next Steps

- Obtain detailed clinical history including fever duration, cough characteristics, dyspnea onset, and previous TB exposure
- Perform vital signs assessment including respiratory rate, oxygen saturation, and blood pressure
- Send sputum samples urgently for TB testing given endemic prevalence in West Africa
- If pneumonia confirmed, repeat CXR in 7-10 days to assess treatment response

Summary: Chest X-ray demonstrates bilateral lower and middle lobe opacities most consistent with pneumonia, though pulmonary tuberculosis and congestive heart failure remain important differential diagnoses. Urgent evaluation with sputum studies, blood cultures, and clinical correlation needed to guide appropriate therapy. Given TB endemicity in West Africa, TB must be excluded before attributing findings to simple pneumonia.

■ **DISCLAIMER:** This report is generated by NexRay AI and is intended solely as a clinical decision-support tool. All findings, conditions, treatments, and recommendations are AI-generated suggestions and do not constitute a formal medical diagnosis or prescription. The clinical judgment of the attending medical professional must be applied before any action is taken.